

(Please write your Enrollment Number)

Enrollment No. _____

MID-TERM EXAMINATION

(Course Name : B. Tech CSE-AI/ECE-AI/AI&ML) (Semester : 1st)

(October 2024) OFF LINE mode

Subject Code: BAS 109	Subject: Applied Mathematics
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Find Eigen values of $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.		CO1
	b) Find the rank of the following matrix $\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 2 & 4 & 5 \end{bmatrix}$		CO1
	c) Find Laplace Transformation of $t \sin t$		CO2
	d) Write the statement of Rank Nullity Theorem		CO2
UNIT I			CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) Find A^{-1} by elementary transformation if $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$		CO1
	b) Test for consistency and solve $5x+3y+7z=4$, $3x+26y+2z=9$, $7x+2y+10z=5$.		CO1
	c) Verify Cayley- Hamilton theorem for $\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$		CO1
UNIT II			CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) Find the Inverse Laplace Transform of $F(s)$, where $F(s) = \frac{1}{s^2+2s+5}$		CO2
	b) Find Half Range Fourier cosine series for x^2 , $0 \leq x \leq \pi$		CO2
	c) If $T:V_2(R) \rightarrow V_3(R)$ defined by $T(a,b) = (a+b, a-b, b)$ is a Linear Transformation then Find rank and nullity of T		CO2